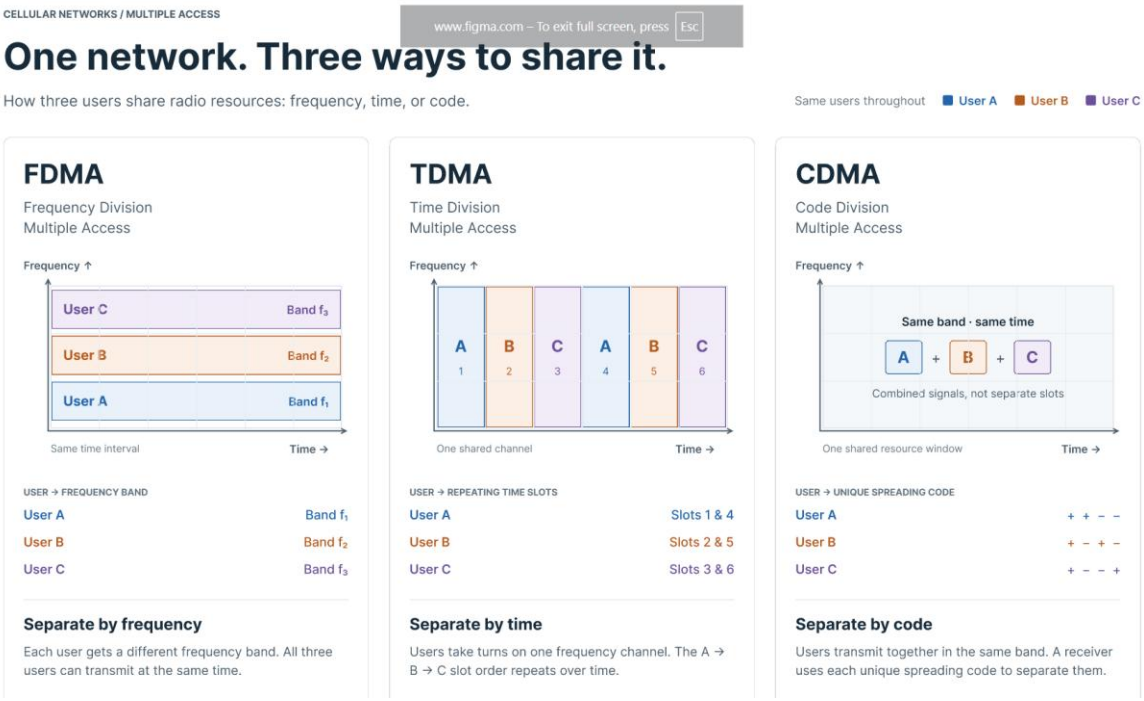
EXPERIMENT 10

Create a visual representation in Figma showcasing the differences between various multiple access techniques.



FDMA

Frequency Division
Multiple Access



USER → FREQUENCY BAND

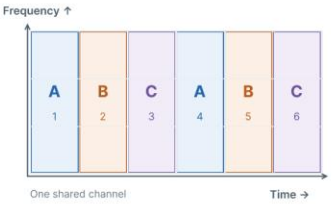
User A	Band f_1
User B	Band f_2
User C	Band f_3

Separate by frequency

Each user gets a different frequency band. All three users can transmit at the same time.

TDMA

Time Division
Multiple Access



USER → REPEATING TIME SLOTS

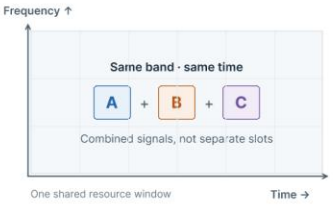
User A	Slots 1 & 4
User B	Slots 2 & 5
User C	Slots 3 & 6

Separate by time

Users take turns on one frequency channel. The A → B → C slot order repeats over time.

CDMA

Code Division
Multiple Access



USER → UNIQUE SPREADING CODE

User A	+ + − −
User B	+ − + −
User C	+ − − +

Separate by code

Users transmit together in the same band. A receiver uses each unique spreading code to separate them.

THE KEY DIFFERENCE

FDMA separates by frequency. TDMA separates by time. CDMA separates by code.

All three allow multiple users to access a network. Diagrams are conceptual; axes and resource windows are consistent, not to scale.